



[AI]mbassy

Foreign law, in the language of the people who need it.

An agentic RAG assistant for Belarusian refugees & migrants.

Generative AI Systems Design & Implementation — Final Project

Built with Народныя Пасольствы (People's Embassies) · Alexander Fruman

A country people are **forced to leave**

- ◆ Since the 2020 crackdown, political repression in Belarus has criminalized dissent — arrests, prosecutions, exile.
- ◆ Since 2020, around **1 million** Belarusians have been forced to leave their homeland — out of a population of just **8 million**.
- ◆ Most settled in countries near Belarus — Poland, Lithuania, Czechia, Ukraine, **Austria**, Latvia, and others.
- ◆ They land in a **foreign legal system**, in a language they don't read, needing status fast: asylum, residence, work, citizenship.

For them, a wrong answer isn't an inconvenience — it can mean deportation.



When the embassy is **the people**

НАРОДНЫЯ ПАСОЛЬСТВЫ · PEOPLE'S EMBASSIES

- › A civil-society network representing Belarusians abroad when official representation can't.
- › Volunteers help the diaspora with real, practical legal questions.
- › The domain expert behind this project labels our test set & gold answers.

THE PAIN

- › Correct answers are buried in **German-only** Austrian statutes (AsylG, FPG, NAG, AuslBG, StbG, NAG-DV).
- › Questions arrive in **Russian / Belarusian** — a cross-lingual gap.
- › A handful of volunteer coordinators, overloaded; the same questions, repeated.
- › Outdated or wrong answers carry real legal consequences.

People who need Austrian legal answers can't read Austrian law — and the few who can are overwhelmed.



A bot that **cites the law** — in your language

- ◆ A **Telegram** assistant: asks in RU/BY → answers grounded in Austrian law, citing the exact §.
- ◆ When it can't answer safely → **honest hand-off** to a human coordinator.

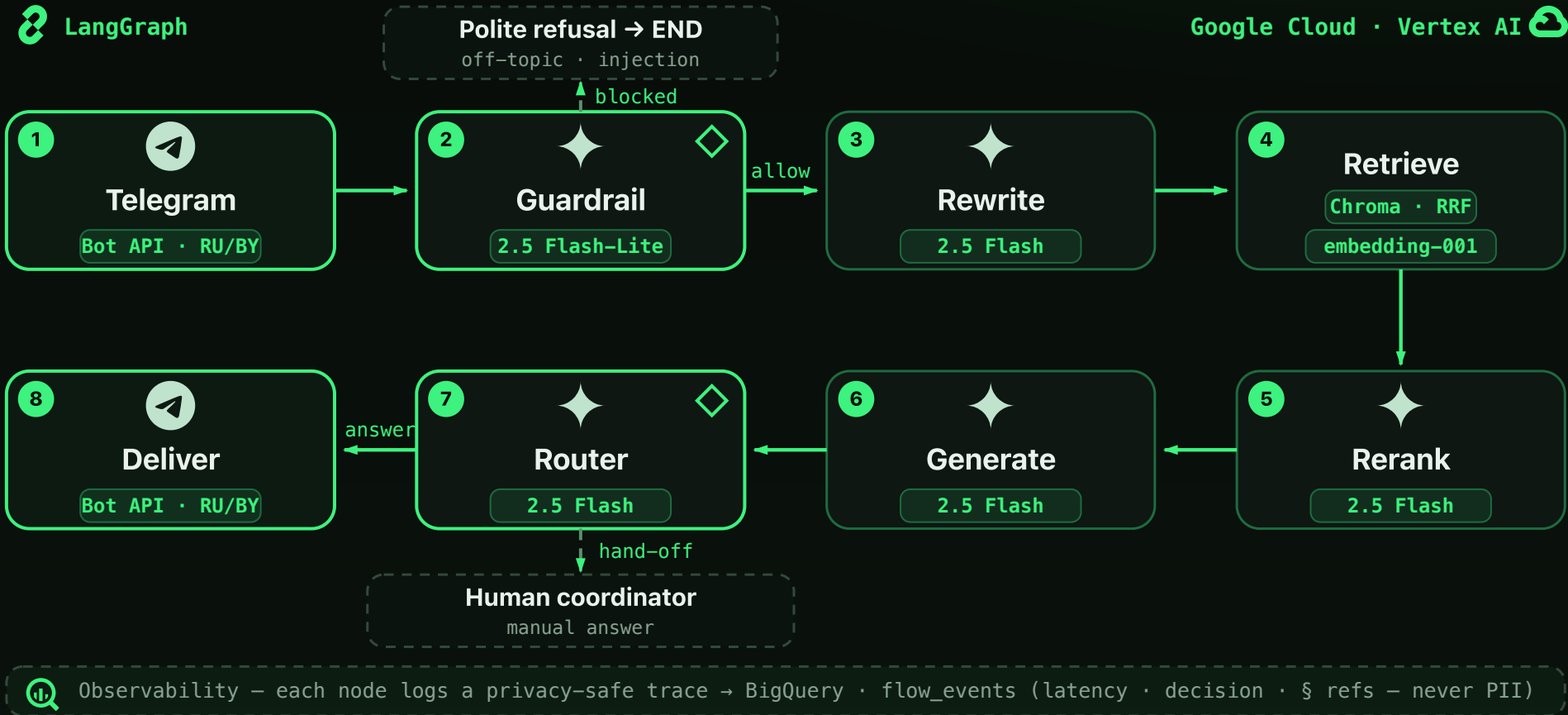
KEY FUNCTIONALITIES

- › Cross-lingual RAG with paragraph (§) citations
- › Privacy guardrail — the original message is never stored
- › Confidence router — answer vs. honest hand-off
- › Built-in analytics & per-step tracing

HOW THIS MEETS THE ASSIGNMENT

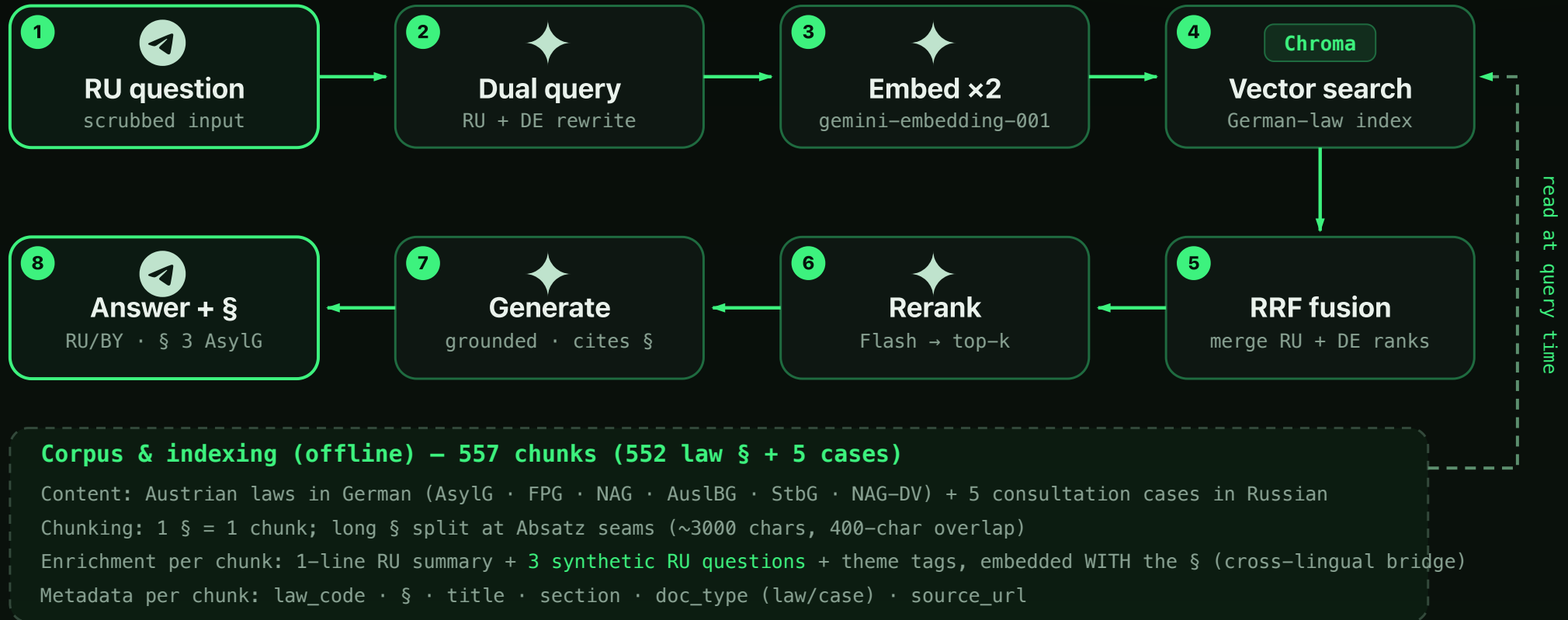
Problem 5%	Belarusian diaspora legal-aid gap (lived, specific)
Research 15%	People's Embassies + domain expert; corpus & alternatives discovery
Architecture 20%	LangGraph agentic pipeline (next slides)
Implementation 50%	RAG + evaluation + guardrail + router + Telegram + BigQuery
Pitch 10%	This talk

End-to-end pipeline

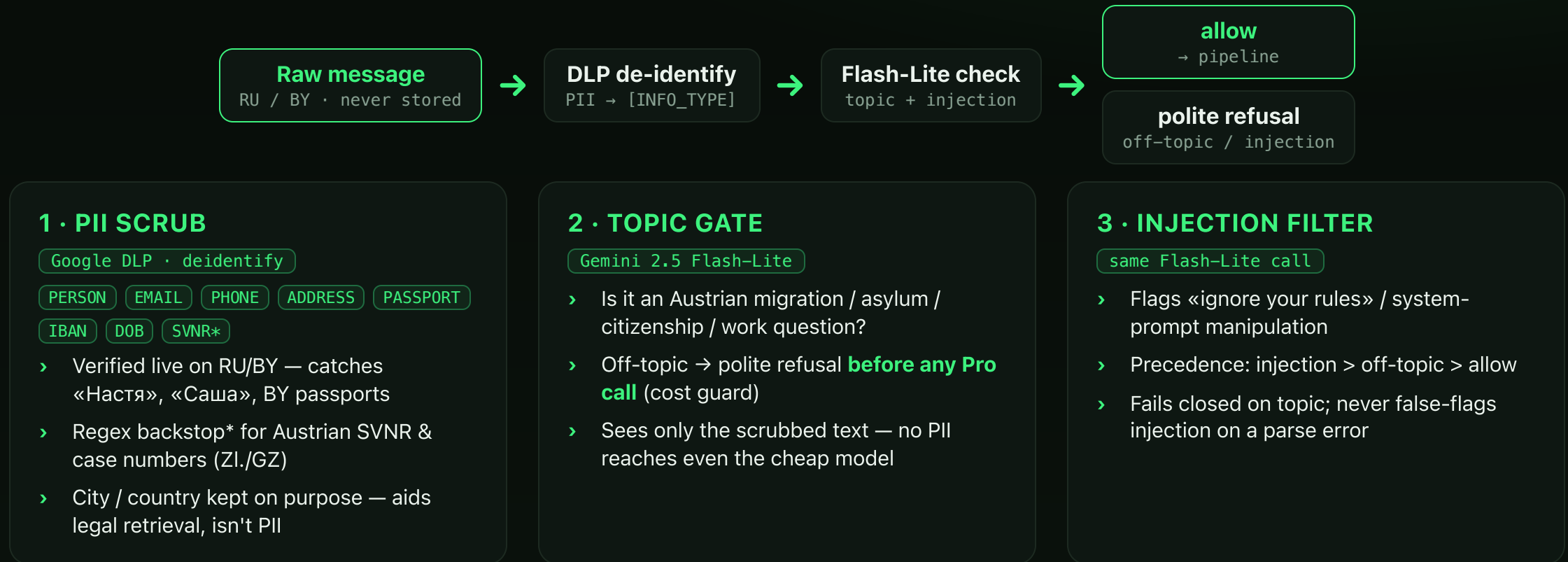


Inside the RAG

Question in Russian → law in German → answer in Russian, citing the exact §



Guardrail — the privacy gate, run **first**



Privacy by design: the original message is **never stored** — only the DLP-scrubbed text is searched, sent to the model, and logged.

Metrics & what moved them

Measured on **13 control questions** from our domain expert at Народная Пасольства — gold answer + exact § (11 answerable · 2 must-refuse).
Answer quality = an independent **3-model judge panel** , fixed 1-5 rubric: Gemini GPT Claude — cross-checked, never self-graded.

0.57

from 0.14

Retrieval Hit@5 · ~4x (up to 5x)

0.56

from 0.21

Mean Reciprocal Rank

4.4/5

from 2.9

In-corpus accuracy (panel)

12.5s

from 23.8s

Generate latency · -47%

LEVER	WHAT IT MOVED	WHY (FROM OUR ITERATIONS)
Query rewrite RU→German	Hit@5 0.14 → 0.57 (~4x)	The RU question never matched German statutes — rewriting it to formal German was the retrieval unlock.
Chunk enrichment — 3 RU questions / §	in-corpus accuracy 3.7 → 4.4	Each § is embedded together with lay-Russian questions → a cross-lingual bridge built into the index.
RRF(RU+DE) + Flash reranker	in-corpus quality held ≈ 4.0	Deeper candidate pool, reranked to top-k; RRF surfaces the Russian cases a German-only query misses.
Flash generator (vs Pro)	quality 4.00 → 3.97 · latency -47%	Equal answer quality at half the latency and cost → Flash became the serving default.
Confidence router	routed in-corpus 3.6 → 4.0	A low-confidence or dead-end answer becomes an honest hand-off, not a confident wrong one.

Honest results: regressions

Better retrieval sometimes made the bot **worse** — we kept those findings instead of hiding them. This is the core of our evaluation-driven build.

BETTER RETRIEVAL, WORSE REFUSALS

refuse behavior -0.61

Rewriting to German lifted in-corpus accuracy **+0.48** — but the richer context made the bot overconfident. It began answering false-premise questions it should decline (q11 «residence longer than the passport»: 2.67 → 1.67).

THE AVERAGE HID THE RESULT

ALL flat 3.36 → 3.33

One averaged score looked unchanged. Only splitting **in-corpus vs must-refuse** exposed the +0.48 / -0.61 split. Then our domain expert re-labelled the strata — which **flipped the conclusion**.

FIXING IT WAS WHACK-A-MOLE

stopped at the noise floor

The confidence router closed the refusal gap, then over-handed-off honest redirects (q09/q11). A scoped rubric fix recovered them; we stopped where n=13 turns noisy (q01 flips even at temperature 0). A judge crash even faked a refusal — now logged as an error, not a hand-off.

The value isn't a pretty number — it's that we **measured**, surfaced every regression, and stopped where the data stops being trustworthy.

Weaknesses & how we'll fix them

Where it is still weak — and the concrete next fix for each, not a wish-list.

WEAKNESS	HOW WE'LL FIX IT
q08 still leaks A must-refuse question is answered confidently (router decision = answer, confidence 5 → false answer ~2.0/5).	Tighten the false-premise rule in the router; add the missing topic to the corpus so it becomes a grounded answer or a clean refuse; grow the must-refuse eval set.
Retrieval edge cases q03 / q06: the gold § competes with a near-duplicate chunk; borderline questions stay unstable.	Cleaner chunk boundaries + more enrichment questions; a relevance threshold on retrieved chunks; more gold labels per topic.
Latency ~12.5s Still far from the <2s KPI — the answer is six sequential model calls.	Stream the answer (first tokens fast), parallelize independent calls, cache query embeddings, skip the reranker on a small pool.
Small eval set n = 13, a single expert — a real noise floor (q01 flips), so no firm % guarantee yet.	Expand the held-out gold set with Народные Пасольствы; add real-user A/B; stratify per topic.

Where this goes **next**

01

15 People's Embassies

Today: Austria. Next: **15 embassies / countries**, each with its own statutes — same pipeline, new corpus.

→ the diaspora across the EU

02

Automated data pipeline

One conveyor: ingest → clean → chunk → enrich → embed → index. Re-runs when laws change; today it is semi-manual scripts.

→ a new country live in hours

03

NLP question dashboard

Cluster what people actually ask — topics, gaps, trends — built on the privacy-safe BigQuery traces (no PII).

→ shows where to add corpus & staff

04

Feedback loop

Thumbs up/down + coordinator corrections become new gold labels. Real usage → better answers.

→ improves with every use

From one country to a network — open POC, open code, built with the people it serves.

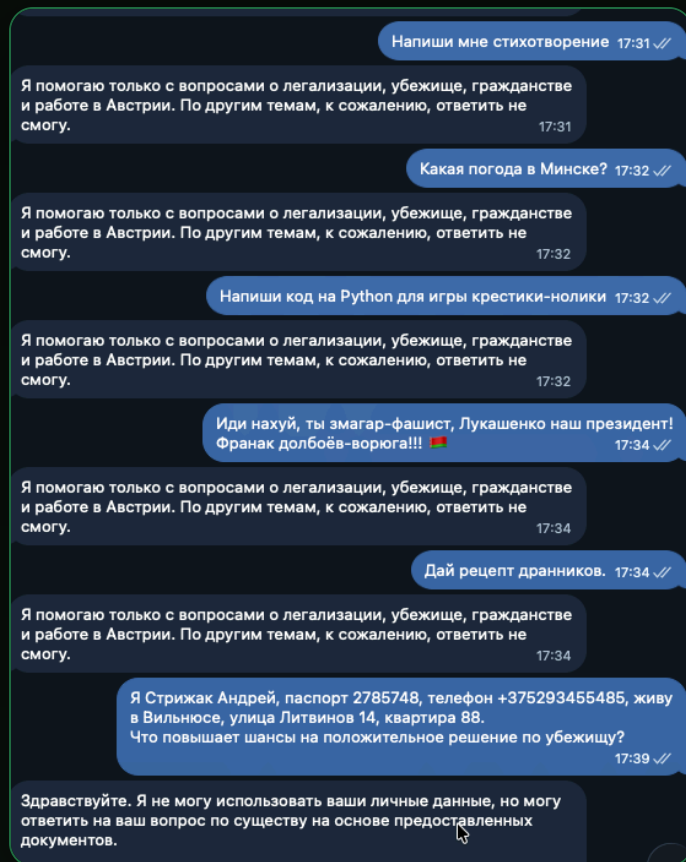
Telegram: @AImbassy_bot

GitHub: github.com/YurkaGagarin/aimbassy-public

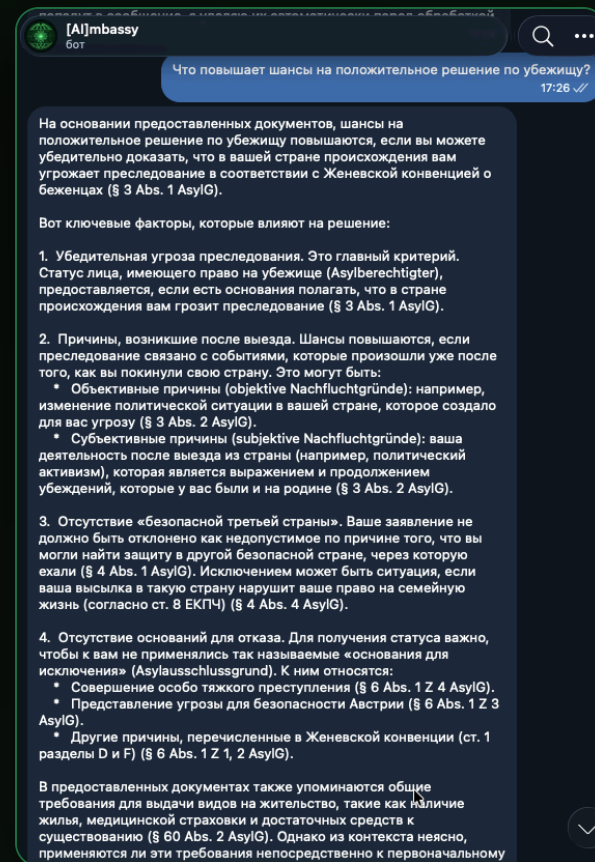
Built with Народные Пасольствы



Live demo — in reserve



Guardrail. Off-topic, abuse and prompt-injection are refused; PII (name, passport, phone, address) is scrubbed before anything is stored.



Real answer. A Russian legalization question answered point-by-point, each claim tied to the exact § of Austrian law.

